FIITJEE
ALL INDIA TEST SERIES
JEE (Advanced)-2023
FULL TEST – III
PAPER –1
TEST DATE: 08-02-2023

Time Allotted: 3 Hours **Maximum Marks: 180**

General Instructions:

- The test consists of total 54 questions.
- Each subject (PCM) has 18 questions.
- This question paper contains **Three Parts.**
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each **Part** is further divided into **Two Sections**: **Section-A & Section-B**.

Section – A *(01 –06, 19 – 24, 37 – 42):* This section contains **EIGHTEEN (18)** questions. Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).

Section – A *(07 – 10, 25 – 28, 43 – 46):* This section contains **TWELVE (12)** Matching List Type Questions. Each question has **FOUR** statements in **List-I** entries (I), (II), (III) and (IV) and **FIVE** statements in **List-II** entries (P), (Q), (R), (S) and (T). The codes for lists have choices (A), (B), (C), (D) out of which, **ONLY ONE** of these four options is correct answer.

Section – B *(11 – 18, 29 – 36, 47 – 54):* This section contains **TWENTY FOUR (24)** numerical based questions. The answer to each question is a **NUMERICAL VALUE.** If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

MARKING SCHEME

Section – A (One or More than One Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+4	If only (all) the correct option(s) is (are) chosen;
Partial Marks	:	+3	If all the four options are correct but ONLY three options are chosen;
Partial marks	:	+2	if three or more options are correct but ONLY two options are chosen and both of which are correct;
Partial Marks	:	+1	If two or more options are correct but ONLY one option is chosen and it is a correct option;
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	–2	In all other cases.

Section – A (Single Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+3	If ONLY the correct option is chosen.
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	–1	In all other cases.

Section – B: Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+3	If ONLY the correct numerical value is entered at the designated place;
Zero Marks	:	0	In all other cases.

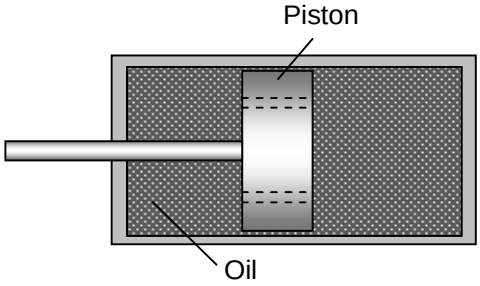
Physics

PART – I

Section – A (Maximum Marks: 24)

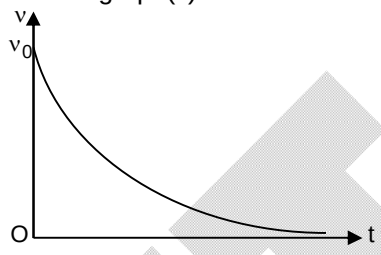
*This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).*

1. The brake mechanism used to reduce recoil in certain types of guns consist essentially of a piston attached to the barrel and moving in a fixed cylinder filled with oil. As the barrel recoils with an initial velocity υ_0, the piston moves and oil is forced through orifices in the piston, causing the piston and the barrel to decelerate at a rate proportional to their velocity; that is, $a = -k\upsilon$, (k > 0), where v and x are the velocity and position of barrel at time t.

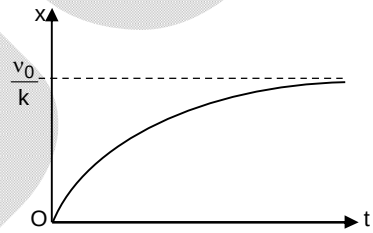


Choose correct graph(s):

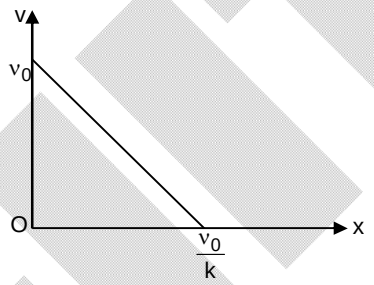
(A)



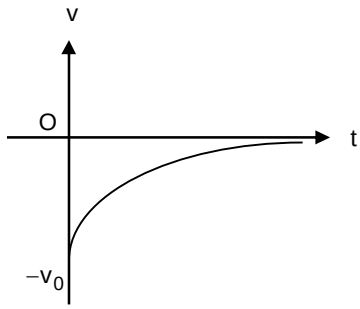
(B)



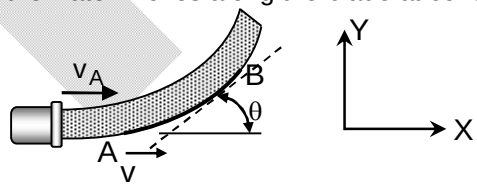
(C)



(D)



2. A nozzle discharges a stream of water of cross – sectional area A with a velocity v_A. The stream is deflected by a single blade which moves to the right with a constant velocity V. Assuming that the water moves along the blade at constant speed, Choose the correct option(s);



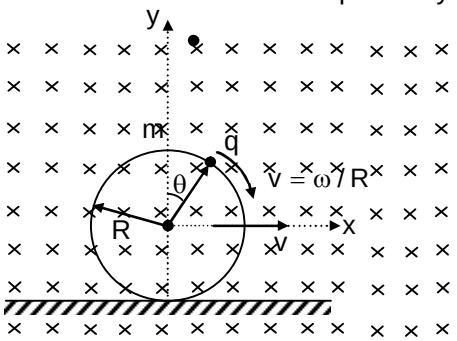
(A) The Y component of the force exerted by the blade on the steam is $A\rho(V_A - V)^2 \sin\theta$.

(B) The X component of the force exerted by the blade on the steam is $A\rho(V_A - V)^2(1-\cos\theta)$

(C) The X component of the force exerted by the blade on the steam is $A\rho(V_A - V)^2\cos\theta$

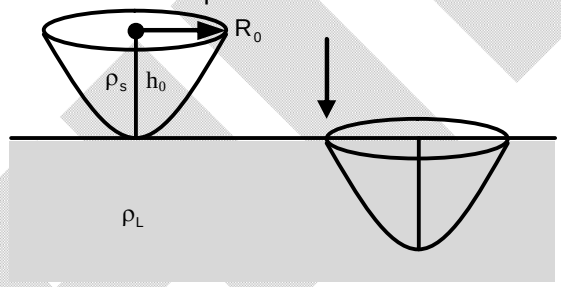
(D) The velocity V for which maximum power is developed is $V = \frac{V_A}{3}$

3. A ring of mass m and radius R is set into pure rolling on horizontal rough surface, in a uniform magnetic field of strength B-as shown in the figure. A point charge(q) of negligible mass is attached to rolling ring. Friction is sufficient so that it does not slip at any point of its motion. (θ is measured in clockwise from positive y-axis)



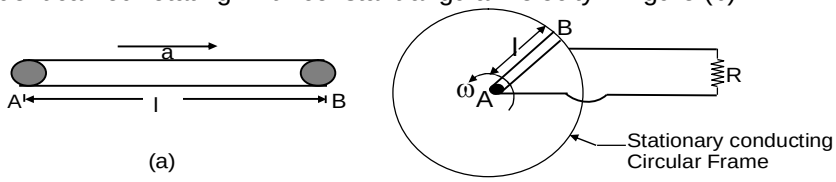
(A) Ring will continue to move with constant velocity.
(B) The value of friction acting on ring is Bqv cos θ.
(C) The value of friction acting on ring is Bqv $\sin\theta$.
(D) Ring will lose contact with ground if v is greater than $\left(\frac{mg}{2qB}\right)$

4. A solid paraboloid of base radius R_0 and height h_0 is having uniform volume mass density of ρ_s is Inverted and placed just above the surface as shown, of a liquid having volume mass density of ρ_L. When paraboloid is released from rest it has been observed that when it becomes completely submerged in the liquid the paraboloid comes to the rest again. There exist uniform gravity of g. Assume that the liquid body is very large so that liquid level is not changing as the paraboloid enters into the liquid.



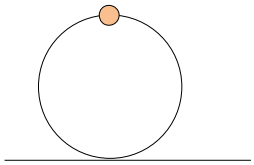
(A) The phenomenon is true for all paraboloids irrespective of their dimensions as long as density ratio is maintained.
(B) The phenomenon is not true for all paraboloids irrespective of their dimensions as long as density ratio is maintained.
(C) The ratio of $\frac{\rho_L}{\rho_s}$ is 2.
(D) The ratio of $\frac{\rho_L}{\rho_s}$ is 3.

5. consider a metal rod of length L that is given a uniform acceleration as shown in figure (a) and an identical rod rotating with constant angular velocity in figure (b).



(A) If V_A and V_B are potentials of end A and B respectively, then $V_A < V_B$ in figure (a)
(B) If V_A and V_B are potentials of end A and B respectively, then $V_A > V_B$ in figure (b)
(C) Electric field inside rod has magnitude $\frac{ma}{e}$ in figure (a)
(D) Electric field inside rod has magnitude $\frac{m\omega^2 r}{e}$ in figure (b)

6. A bead is fixed on a rigid ring of radius r and negligible mass as compared to that of the bead. Assembly thus formed is held motionless on a horizontal floor with the plane of the ring vertical and the bead on the top. Coefficient of friction between the ring and the floor is μ. Now the assembly is given a slight push so that the ring starts rolling Acceleration of free fall is g. Choose the correct statements.



(A) The angle rotated by the ring before it begins to slide is $\tan^{-1}(\mu)$
(B) The speed of centre of the ring before it begins to slide is $\mu\sqrt{gR}$
(C) The angle rotated by the ring before it begins to slide is $2\tan^{-1}(\mu)$
(D) The speed of the centre of the ring before it begins to slide is $\mu\sqrt{2gR}$

Section – A (Maximum Marks: 12)

*This section contains **FOUR (04)** Matching List Type Questions. Each question has FOUR statements in **List-I** entries (I), (II), (III) and (IV) and FIVE statements in **List-II** entries (P), (Q), (R), (S) and (T). The codes for lists have choices (A), (B), (C), (D) out of which **ONLY ONE** of these four options is correct answer.*

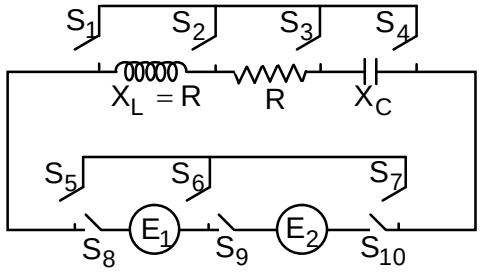
7. The ground state energy of hydrogen atom is -13.6 eV. A sample of hydrogen atoms in its ground state is irradiated with photons of energy 10.2 eV. The radiations from the above sample is then incident on two sample containing excited He^+ ions and excited Li^{2+} ions respectively. Both the samples absorb the radiations. (use $hc = 12400\ eV\,\overset{o}{A}$)

List-I		List-II	
(I)	No. of spectral lines obtained in the Spectra of Li^{2+}	(P)	6
(II)	No. of spectral lines obtained in the Spectra of He^+	(Q)	15
(III)	Smallest wavelength in spectra of He^+ is $(48.8)X\ \overset{o}{A}$ then X (Approx.) is	(R)	5
(IV)	Smallest wavelength in spectra of Li^{2+} is $(10.4)Y\ \overset{o}{A}$ then Y (Approx.) is	(S)	10
		(T)	18

Codes:

	I	II	III	IV
(A)	R	P	P	T
(B)	Q	P	P	Q
(C)	Q	P	R	S
(D)	T	P	R	Q

8. Consider a given series LCR circuit as shown in figure. There are 10 switches. In list-I switches which are closed, is given and corresponding value is asked. In list-II these values are given. Choose the proper value from list-II to list-I using the codes given below.



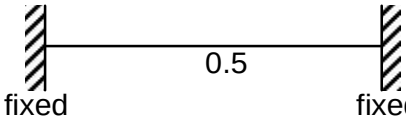
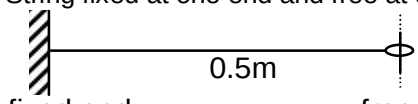
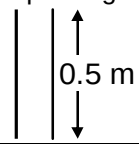
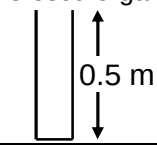
$E_1 = E_0 \sin\omega t \qquad E_2 = E_0 \sin(\omega t + \phi)$

List-I		List-II	
(I)	It is observed that effective current in the circuit in case if either S_8, S_6 and S_7 or if S_5, S_6, S_9, S_{10} or if S_8, S_9 and S_{10} are closed (Rest open) remains same. Value of ϕ is:	(P)	$\frac{2\pi}{5}$
(II)	Switch S_8, S_9 & S_{10} are closed. Now its observed if rest switches are opened or switches S_3, S_4 are closed (rest opened) effective value of current in the circuit is same, value of $\frac{4\pi X_C}{5R}$ is	(Q)	$\frac{\pi}{2}$
(III)	Switch S_8, S_6, S_7 are closed. Now effective current if S_1, S_2 closed (Rest opened) is I_1 and If S_3, S_4 closed (Rest opened) is I_2, value of $\pi\left(\frac{I_1}{I_2}\right)^2$ is	(R)	$\frac{2\pi}{3}$
(IV)	Switch S_5, S_6, S_9 and S_{10} are closed. Peak value current if rest are opened is I_3 and if S_2, S_3 closed (Rest opened) is I_4. Value of $\pi\left(\frac{I_3}{I_4}\right)^2$ is	(S)	$\frac{8\pi}{5}$
		(T)	$\frac{\pi}{3}$

Codes:

	I	II	III	IV
(A)	Q	S	P	T
(B)	Q	T	P	S
(C)	R	P	T	Q
(D)	R	S	P	Q

9. In each of the four situations of List I, a stretched string or an organ pipe is given along with the required data. In case of strings the tension in strings is $T = 102.4$ N and the mass per unit length of string is 1 g/m. Speed of sound in air is 320 m/s. Neglect end corrections. The frequencies of resonance are given in List II. Match each situation in List I with the possible resonance frequencies given in List II.

List-I		List-II	
(I)	String fixed at both ends 	(P)	320 Hz
(II)	String fixed at one end and free at other end 	(Q)	480 Hz
(III)	Open organ pipe 	(R)	640 Hz
(IV)	Closed organ pipe 	(S)	800 Hz
		(T)	1280 Hz

Codes:

	I	II	III	IV
(A)	P	R	Q	S
(B)	P	S	R	Q
(C)	R	Q	S	P
(D)	Q	S	P	R

10. There exists a uniform electric field in space. Potential of few points in space is given below

Coordinates	(0, 0, 0)	(a, 0, 0)	(0, b, 0)	(0, 0, c)
Potential	1 V	2 V	3 V	4 V

Match the points in List I with their potential in List II

List-I		List-II	
(I)	(a, b, 0)	(P)	4V
(II)	(a, b, c)	(Q)	7V
(III)	(-a, -b, -c)	(R)	0V
(IV)	(0, b, c)	(S)	6V
		(T)	–5V

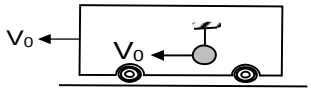
Codes:

	I	II	III	IV
(A)	R	T	Q	S
(B)	P	Q	T	S
(C)	S	Q	P	R
(D)	T	P	Q	R

Section – B (Maximum Marks: 24)

*This section contains **EIGHT (08)** numerical based questions. The answer to each question is a **NUMERICAL VALUE.** If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.*

11. A pendulum with massless string of length l and bob of mass m is attached inside a railway carriage which is moving with speed v_0 as shown. Carriage is brought to rest uniformly by applying brakes. When carriage comes to rest pendulum string is horizontal.
Minimum magnitude of retardation of carriage such that bob completes vertical circular motion is (in ms^{-2}).
($g = 10\ ms^{-2}$)



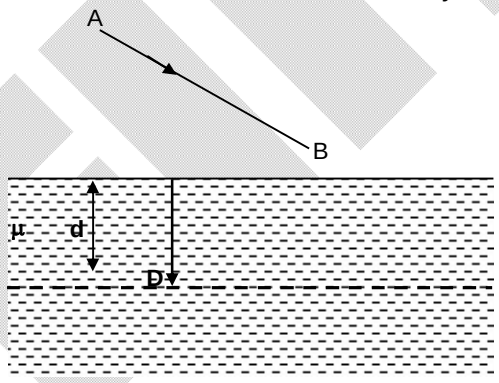
12. A small boat is travelling downstream with velocity 5 m/s relative to river flowing with speed 7/3 m/s. it encounters a log floating in the river at some distance from itself. It turns by an angle of 37° and just crosses the edge of the log after 8 sec. Find time (in seconds) taken by boat to just cross the edge of log if log is kept fixed.



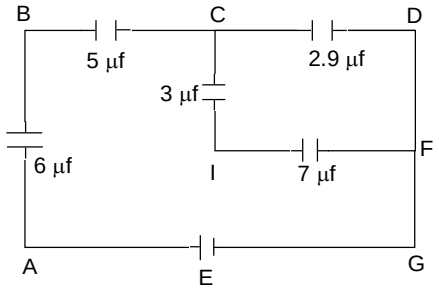
13. In YDSE, if sources are incoherent, the intensity on screen is $13I_0$. When these sources are coherent then minimum intensity on screen is I_0. The maximum intensity produced by these coherent sources on screen is___W/m^2 $\left[\text{given } I_0 = 1 W/m^2\right]$

14. The electrostatic flux through a rectangular area of $\frac{7}{188} m^2$ lying in a plane $2x + 3y + 6z = 10$ due to an electric field $\left(8\hat{i} + 6\hat{j} + 10\hat{k}\right) V/m$ will be in (V – m)

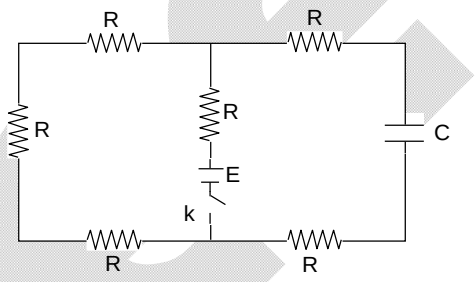
15. A diver D is still under water $\left(\mu = \frac{4}{3}\right)$ at a depth d = 10m. A bird is diving along line AB at a constant velocity in air. When the bird is exactly above the diver he sees it at a height of 50 m from himself and velocity of the bird appears to be inclined at 45^0 to the horizontal. At what distance from the diver, the bird actually hits the water surface. (in meter)



16. Consider the circuit shown. If in steady state $V_A - V_C = 1$ volt, find $11(V_I - V_G)$ in steady state.



17. In the given circuit initially the capacitor is uncharged. At t = 0 the key k is closed. The time constant of circuit is n times of RC. Find the value of 'n'.



18. The pitch of a screw gauge is 1 mm and its cap is divided into 100 divisions. When nothing is placed between its studs, the zero of the circular scale lies 6 divisions below the reference line. When a wire is placed between its studs, the main scale reading is 4 mm and 56th division of circular scale coincides with the reference line. The length of the wire is 4.20 cm. Find the curved surface area (*in* cm^2) of the wire in true significant figures.

Chemistry

PART – II

Section – A (Maximum Marks: 24)

*This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).*

19. The radial wave equation for hydrogen atom is:

$$\psi = \frac{1}{16\sqrt{4}}\left(\frac{1}{a_0}\right)^{3/2}\left[(x-1)\left(x^2-8x+12\right)\right]e^{-x/2}$$

where $x = 2r/a_0$, $a_0 =$ radius of first Bohr's orbit.

Select the correct statement(s)

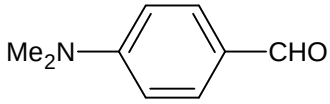
(A) minimum position of node will be $0.5a_0$

(B) maximum position of node will be $3a_0$

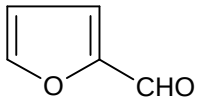
(C) There will be 3 radial nodes

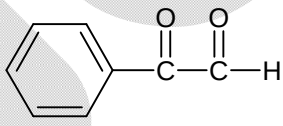
(D) There will be 2 radial nodes

20. How many of the following will give Cannizzaro reaction?

(A) 

(B) 

(C) 

(D) 

21. On the basis of MO theory, which is correct?

(A) The bond order for C_2 molecule is two and both bond are π bond

(B) The bond order for C_2 molecule is two and one bond is π and one is σ

(C) The bond order for C_2 molecule is two and both bond are σ bond

(D) Total bonding e^- in C_2 is 8

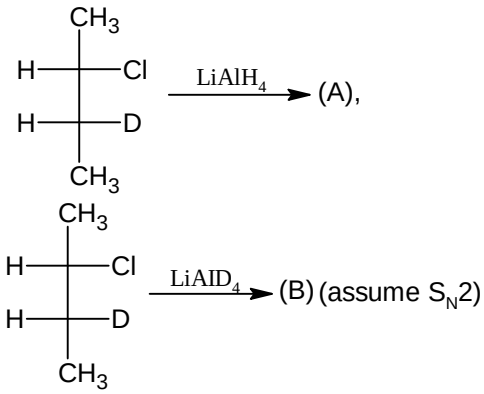
22. Which of the following statement(s) is (are) correct about slag?

(A) The calcium silicate, $CaSiO_3$ is obtained in slag formation zone in the extraction of iron from haematite ore.

(B) The chemical composition of "slag" formed during the smelting process in the extraction of copper is $FeSiO_3$

(C) In blast furnace/Bessemer converter, the upper layer of molten liquid (i.e. molten metal) is of slag.

(D) The slag is fusible matter.

23. 

Select the correct is/are statement(s)
(A) A is optically active
(B) B is optically active
(C) A is optically active, but B is optically inactive
(D) A is optically inactive, B is optically active

24. Which of the following orders is/are correct?
(A) $SbH_3 > NH_3 > AsH_3 > PH_3$ – Boiling point
(B) $NH_3 > PH_3 > AsH_3 > SbH_3$ – Thermal stability
(C) $NH_3 > PH_3 > AsH_3 > SbH_3$ – Basic character
(D) $NH_3 > PH_3 > AsH_3 > SbH_3$ – Bond Angle

Section – A (Maximum Marks: 12)

*This section contains **FOUR (04)** Matching List Type Questions. Each question has FOUR statements in **List-I** entries (I), (II), (III) and (IV) and FIVE statements in **List-II** entries (P), (Q), (R), (S) and (T). The codes for lists have choices (A), (B), (C), (D) out of which **ONLY ONE** of these four options is correct answer.*

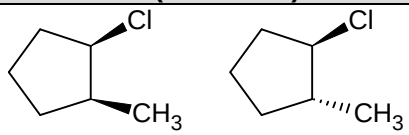
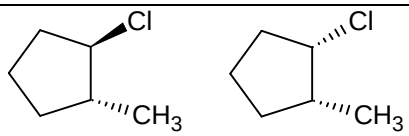
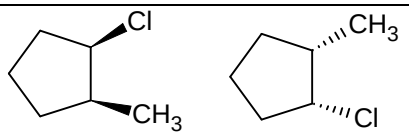
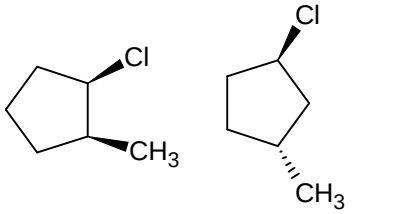
25. Consider the following list:

List - I			List - II		
	Electrode			Anode	Cathode
(I)	Fused NaCl	Graphite & mercury	(P)	$O_{2(g)}$	$Na(g)$
(II)	Conc. NaCl	Pt	(Q)	$O_{2(g)}$	$H_2(g)$
(III)	Dil. NaCl	Pt	(R)	$Cl_{2(g)}$	$H_2(g)$
(IV)	Dil. $NaCl_{(aq)}$	Mercury	(S)	$Cl_{2(g)}$	$Na_{(s)}$
			(T)	$O_{2(g)}$	$Cl_{2(g)}$

Codes:

	I	II	III	IV
(A)	S	R	Q	P
(B)	T	R	Q	P
(C)	S	T	Q	P
(D)	S	R	T	P

26. Match List-I with List-II:

List – I (Molecules)		List – II (Relationship)	
(I)		(P)	Identical
(II)		(Q)	Enantiomer
(III)		(R)	Diastereomer
(IV)		(S)	Structural isomerism
		(T)	Homomers

Codes:

	I	II	III	IV
(A)	R	T	P	S
(B)	R	R	T	S
(C)	R	R	P	T
(D)	R	R	P	S

27. Consider the following lists:

List - I		List - II	
(I)	$CH_2 = CH - CHO \xrightarrow{NaBH_4,alc}$	(P)	Acetal
(II)	$C_6H_5CHO + Ph - NH_2 \xrightarrow{pH(4-5)}$	(Q)	Schiff's base
(III)	$C_6H_5COCH_3 + CH_3 - CH_2 - NH - CH_3 \xrightarrow{pH(4-5)}$	(R)	Unsaturated alcohol
(IV)	$RCHO + 2RCH_2OH \xrightarrow{H^+}$	(S)	Enamine
		(T)	Imine

Which one is correct?

Codes:

	I	II	III	IV
(A)	R	Q	S	P
(B)	T	Q	S	P
(C)	R	T	S	P
(D)	R	Q	T	P

28. Consider the following lists:

List - I		List - II	
(I)	$Me_3C-Cl+EtO^-Na^+$	(P)	Me_3C-OEt
(II)	$EtCl+Me_3CO^-Na^+$	(Q)	Alcohol
(III)	$PhCl+MeO^-Na^+$	(R)	Anisole
(IV)	PhO^-Na^++MeCl	(S)	No result
		(T)	$Ph-O-Ph$

Which one is correct?

Codes:

	I	II	III	IV
(A)	P	Q	S	R
(B)	S	P	Q	R
(C)	Q	P	S	R
(D)	Q	P	T	R

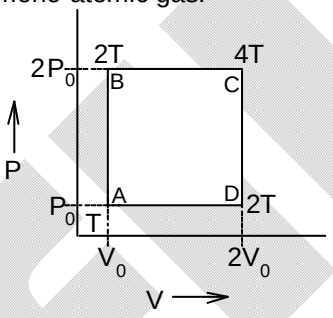
Section – B (Maximum Marks: 24)

*This section contains **EIGHT (08)** numerical based questions. The answer to each question is a **NUMERICAL VALUE.** If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.*

29. With strong oxidizing agent, the oxidation product of $K_4\left[Fe(CN)_6\right]$ is K_2O, Fe_2O_3, NO_3^- and CO_3^{2-}, then what will be the 'n' factor of $K_4\left[Fe(CN)_6\right]$.

30. For the acid indicator (HIn) thymol blue, pH is 2.176 when 60% of the indicator is in ionized form. Find the % indicator in unionized form in the solution with $\left[H^+\right]=3.33\times10^{-3}$M.

31. Find out the efficiency (in %) of the following cyclic process $A\to B\to C\to D\to A$ for 1-mole mono-atomic gas.



32. A solution having a vapour pressure of 520 mm of Hg contains o-nitrophenol and p-nitrophenol with mole fraction of 0.6 and 0.4 respectively. One of the above has $P^\circ=400$ mm of Hg. The vapour pressure of o-nitrophenol is

33. A hydrogen like species with atomic number 'Z' is in excited state 'n' and emits photons of energy 25.7 and 8.7 eV when makes a transitions to 1st and 2nd excited state respectively. Ionisation energy (eV) of specie is

34. What should be the relative humidity below which $CuSO_4.5H_2O$ would efflorescent at $25^\circ C$? For the reaction

$$CuSO_4.5H_2O_{(s)} \rightleftharpoons CuSO_4.3H_2O_{(s)} + 2H_2O_{(g)}$$

at $25^\circ C$ Kp is $1.0 \times 10^{-4} atm^2$ and vapour pressure of water is 30.4 torr. (Given: Relative humidity is the ratio of amount moisture in air to the maximum amount of moisture that could exist in air at specific temperature)

35. Potassium has a bcc structure with nearest neighbours distance 4.52 A. Its atomic weight is 39. Its density (in kg m^{-3}) will be:

36. Two moles of an ideal triatomic gas $\left(\gamma = \frac{4}{3}\right)$ at 16 bar and 300 K are expanded reversibly upto a final pressure of 1 bar. How much work (in J (magnitude)) is done by the gas?

Mathematics

PART – III

Section – A (Maximum Marks: 24)

*This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).*

37. If $\alpha_1, \alpha_2, \alpha_3, \alpha_4, \alpha_5$ are roots of equation $x^5 - 5x^4 - 1 = 0$, then

(A) $\sum_{r=1}^{r=5} \frac{1}{\alpha_r^4} = -\frac{1}{20}$ (B) $\sum_{r=1}^{r=5} \frac{1}{\alpha_r^4} = -20$

(C) $\prod_{r=1}^{r=5} \left(\frac{1}{\alpha_r^4} + 5\right)^5 = 1$ (D) $\prod_{r=1}^{r=5} \left(\frac{1}{\alpha_r^4} + 5\right)^3 = \frac{1}{5}$

38. If three non-zero and non-collinear vectors $\hat{i} + a\hat{j} - \hat{k}$, $2\hat{i} - \hat{j} + 3\hat{k}$ and $a\hat{i} - \hat{j} + 3\hat{k}$ are coplanar, then 'a' is/are

(A) $\frac{1}{3}$ (B) 2

(C) $-\frac{1}{3}$ (D) –2

39. If $3f(x) + 4f(3-x) = 3x^2 - 8$ and $\int_{\alpha}^{\beta} f(x)dx \le 0$, where $\alpha, \beta\ (\alpha < \beta)$ are roots of $x^2 - 3x + a = 0$, then 'a' can be

(A) $\frac{-1}{101}$ (B) $\frac{1}{101}$

(C) $\frac{2}{301}$ (D) $\frac{401}{301}$

40. If the curves $y = \lambda x^2 + \lambda x + \frac{1}{24}$ and $x = \lambda y^2 + \lambda y + \frac{1}{24}$ are tangent to each other, then λ can be

(A) $\frac{2}{3}$ (B) $\frac{3}{2}$

(C) $\frac{13 + \sqrt{601}}{12}$ (D) $\frac{-13 + \sqrt{601}}{12}$

41. $\lim_{n\to\infty} \frac{1}{n^2}\left[(n+1)\left(n+\frac{1}{2}\right)\left(n+\frac{1}{2^2}\right).....\left(n+\frac{1}{2^{n-1}}\right)\right]^n$ is 'a', then

(A) $[a] = 5$ (B) $[a] = 7$

(C) $(a) = 6$ (D) $(a) = 8$

Here (.) represents least integer part and [.] represents greatest integer part.

42. a, b, c are distinct complex numbers with $|a| = |b| = |c| = 1$ and z_1, z_2 be the roots of the equation $az^2 + bz + c = 0$ with $|z_1| = 1$. Also, let P and Q represents the complex numbers z_1 and z_2 in the complex plane with $\angle POQ = \theta$, where O is origin. Then,

(A) $b^2 = ac$, $\theta = \frac{2\pi}{3}$ (B) $\theta = \frac{2\pi}{3}$, $PQ = \sqrt{3}$

(C) $b^2 = ac$, $PQ = 2\sqrt{3}$ (D) $2b^2 = ac$, $\theta = \frac{\pi}{3}$

Section – A (Maximum Marks: 12)

*This section contains **FOUR (04)** Matching List Type Questions. Each question has FOUR statements in **List-I** entries (I), (II), (III) and (IV) and FIVE statements in **List-II** entries (P), (Q), (R), (S) and (T). The codes for lists have choices (A), (B), (C), (D) out of which **ONLY ONE** of these four options is correct answer.*

43. Let N be the number of words which can be formed using all the letters of the word "DARJEELING" so that there are at least two consonants between any two vowels. Then match List I with List II:

List - I		List - II	
(I)	If N is divisible by 2^n $(n \in N)$, then n can be	(P)	1
(II)	If N is divisible by 6^P $(P \in N)$, then P must be less than	(Q)	2
(III)	Number of odd divisors of N is greater than	(R)	3
(IV)	Number of zeroes at the end of N is less than	(S)	4
		(T)	5

Codes:

	I	II	III	IV
(A)	PQR	QRST	PQRST	PST
(B)	QRT	PQT	PQRS	PQRST
(C)	PQRST	ST	PQRST	QRST
(D)	PQST	PQRST	PQRT	PQR

44. Given $y = ax^2 + bx + c$ is a quadratic expression for $x \in [0, 2]$, y falls completely in [0, 1] and $-\frac{b}{2a} \notin (0, 2)$. Then, for $a \neq 0$, match the following:

List - I		List - II	
(I)	The value of c may be	(P)	0
(II)	The value of $\lvert 2a + b \rvert$ may be	(Q)	1
(III)	If $\frac{\lambda}{a}$ is a root of $ax^2 + bx + c = 0$, then λ may be	(R)	$\frac{1}{2}$
(IV)	The sum of roots of $ax^2 + bx + c = 0$ is $(p^2 + p + 3)$ for $p \in R$, then the value of $\lvert a \rvert$ can't be	(S)	$\frac{4}{3}$
		(T)	2

Codes:

	I	II	III	IV
(A)	PQ	R	PR	QST
(B)	PQR	PR	T	ST
(C)	ST	Q	PQ	RS
(D)	R	ST	QST	PQ

45. Match the equations in List I to solution in List II:

List - I		List - II	
(I)	$\frac{2}{\pi}\tan\left(\frac{\pi x}{2}\right)+\frac{1}{2x}=\left(\frac{\pi+2}{\pi}\right)e^{-k}$ for $0 < x < 1$	(P)	Only one solution for $k \in (0,1/e)$
(II)	$\ln x = kx$ has	(Q)	Only one solution for $k \in \left(-\frac{1}{e},0\right)$
(III)	$xe^x = k$ has	(R)	Two solutions for $k \in \left(\frac{-1}{e},0\right)$
(IV)	$x\ln x = k$ has	(S)	Two distinct solutions for $k \in \left(0,\frac{1}{e}\right)$
		(T)	No solution for $k \in \left(0,\frac{1}{e}\right)$

Codes:

	I	II	III	IV
(A)	PQ	T	S	PR
(B)	T	PT	PR	QS
(C)	QS	Q	PT	R
(D)	RT	QS	PR	PR

46. Match the locus of 'z' in List I from List II:

List - I		List - II	
(I)	Straight line	(P)	If $z^2\bar{z}+\bar{z}^2z=(z+\bar{z})$; then z lies on
(II)	Circle	(Q)	If $\|\|z-3|-|z+3\|\| \le 6$, then z lies on
(III)	Parabola	(R)	If $z^3\bar{z}+z\bar{z}^3-2\left(z^2+\bar{z}^2-4(|z|^2-2)\right)$, then z lies on
(IV)	Hyperbola	(S)	$\left(z^2+\bar{z}^2\right)-4\,\text{Re}(z)\times\left((z-\bar{z})^2-(z+\bar{z})\right)=0$, then z lies on
		(T)	$z^2+\bar{z}^2=z\bar{z}$, then z lies on

Codes:

	I	II	III	IV
(A)	PQ	PR	S	QR
(B)	Q	PR	QS	T
(C)	QR	QR	S	P
(D)	P	T	Q	S

Section – B (Maximum Marks: 24)

*This section contains **EIGHT (08)** numerical based questions. The answer to each question is a **NUMERICAL VALUE.** If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.*

47. If $\cos^{-1}(2-x^2)+\sin^{-1}(2-x^2)+\tan^{-1}x \ge \frac{2\pi}{3}$. Given the solution set, $\alpha \le x \le \beta$, then $\left(\alpha+\frac{1}{\beta^2}\right)$ is equal to

48. If $y=\dfrac{x}{x^2+\dfrac{x}{x^2+\ldots\infty}}$, then $\int\dfrac{(y-x^2)dx}{(x^2+y)(x+y^2)}=f(y)$. Then, $f'(4)$ is equal to

49. If $(9\sin x+4\,\text{cosec}\,x)=2k+4$ has only three solutions, then $k=$

50. If $\sqrt{5}x - 2y + 5 = 0$ is a tangent at any point P on the parabola whose focus is $\equiv \left(-\sqrt{5}, 2\right)$ & length of latus rectum is 4 units; intersect axis at A & normal at point P intersect axis at B, then the circum radius of ΔPAB is

51. If $\frac{1}{e-1}\sum_{n=1}^{50}\frac{1}{(n+1)!}\int_{-n}^{n} e^{\{|x|\}}\frac{dx}{1+e^{\sin x}} = 1 - \frac{1}{(ab)!}$
where a and b are prime, then $|a-b|$ is equals to

52. Let $f(x) = x^3 - x^2 - x + 1, g(x) = a^2 - 2$ (where a is real parameter) and $f(x) = g(x)$ has three real and distinct roots, then $f(x) = [a^2]$ has real root having the integral part (where [.] is G.I.F) equal to

53. If $\frac{3x^2+1}{\sqrt{x^4+x^2}} \geq \max f(x)$, where f(x) is constant for $0 < x \leq \sqrt{2}$, then minimum value of
$\int_0^{\sqrt{2}} \frac{3x^2+1}{\sqrt{x^4+x^2}}dx$ is

54. Let x, y, z are selected from the set of natural numbers. If the probability that $x^2 + y^2 + z^2$ is divisible by 5 is $\frac{p}{q}$ (where p, q are coprime), then p + q is equal to